

5.5.25

AI25BTECH11008 - Chiruvella Harshith Sharan

Question:

If

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 0 & -3 \\ 1 & 2 & 0 \end{pmatrix}$$

then find A^{-1} and use it to solve the system

$$x + 2y - 3z = 1, \quad 2x - 3z = 2, \quad x + 2y = 3.$$

Solution:

Step 1: Matrix form

Write the system as a matrix equation

$$A\mathbf{x} = \mathbf{b}. \quad (1)$$

Here

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}. \quad (2)$$

Step 2: Compute $\det(A)$ and the adjugate

We compute the determinant by expansion (first row):

$$\det(A) = 1 \cdot \det \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix} - 2 \cdot \det \begin{pmatrix} 2 & -3 \\ 1 & 0 \end{pmatrix} - 3 \cdot \det \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} = 1 \cdot 6 - 2 \cdot 3 - 3 \cdot 4 = -12.$$

Next compute the matrix of cofactors (each entry is the signed minor). The cofactor matrix C is

$$C = \begin{pmatrix} 6 & -3 & 4 \\ -6 & 3 & 0 \\ -6 & -3 & -4 \end{pmatrix}.$$

The adjugate is the transpose of C :

$$\text{adj}(A) = C^T = \begin{pmatrix} 6 & -6 & -6 \\ -3 & 3 & -3 \\ 4 & 0 & -4 \end{pmatrix}. \quad (3)$$

Step 3: Inverse from adjugate and determinant

Using $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ we get

$$A^{-1} = \frac{1}{-12} \begin{pmatrix} 6 & -6 & -6 \\ -3 & 3 & -3 \\ 4 & 0 & -4 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \\ -\frac{1}{3} & 0 & \frac{1}{3} \end{pmatrix}. \quad (4)$$

Step 4: Solve $\mathbf{x} = A^{-1}\mathbf{b}$

Multiply to obtain the solution:

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \\ -\frac{1}{3} & 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{1}{2} \\ \frac{2}{3} \end{pmatrix}.$$

Final Answer:

$$\boxed{x = 2, \quad y = \frac{1}{2}, \quad z = \frac{2}{3}}$$

Below is a schematic TikZ illustration projecting the first two components of the vectors (a projection to the xy -plane for visualization). The algebraic relation is exact in $\mathbb{R}^3$; the drawing is only a 2D aid.

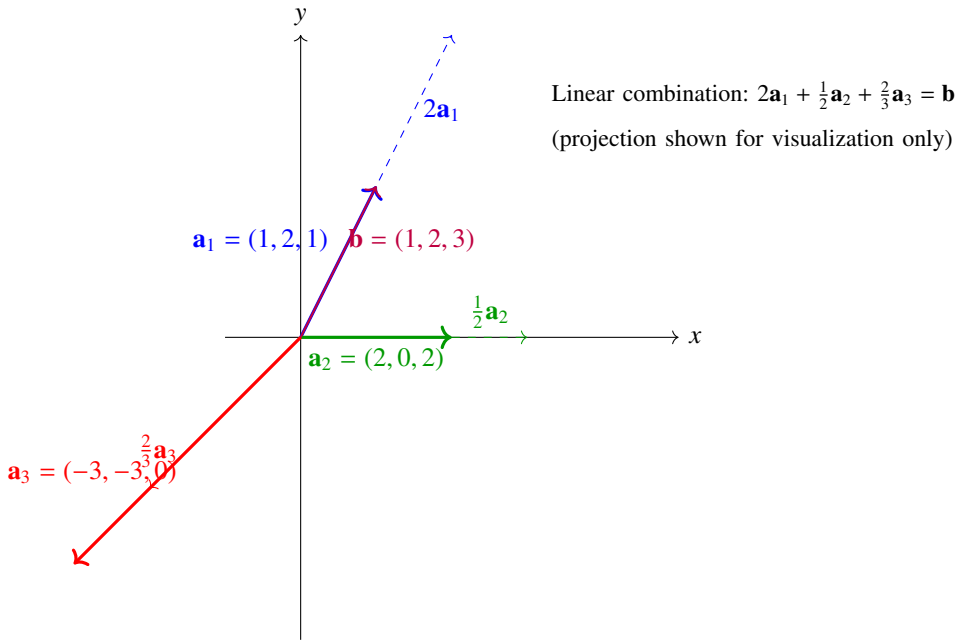


Fig. 1: Schematic (projection) showing column vectors of A and the right-hand side $\mathbf{b}$.